



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis (MAT221)

Fundamental Theorems of Calculus

FTC-1, FTC-2, Average Value Theorem

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CONDUCTED BY

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History of Calculus

Fundamental Theorem of Calculus

💡 Fundamental Theorem of Calculus Part 1

Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable, and define

$$F(x) = \int_a^x f(t) dt \quad \text{for all } x \in [a, b].$$

Then F is continuous on $[a, b]$. If f is continuous at some point $c \in [a, b]$, then F is differentiable at c and

$$F'(c) = f(c).$$



💡 Fundamental Theorem of Calculus Part 2

If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, and $F : [a, b] \rightarrow \mathbb{R}$ satisfies

$$F'(x) = f(x) \quad \text{for all } x \in [a, b],$$

then

$$\int_a^b f = F(b) - F(a).$$



Some Consequences of the Fundamental Theorem of Calculus

Theorem

We have seen that not every derivative is continuous, but explain how we at least know that every continuous function is a derivative. 

Theorem

Let

$$f(x) = |x|$$

and define

$$F(x) = \int_{-1}^x f(t) dt.$$

1. Find a formula for $F(x)$ for all x . Where is F continuous? Where is F differentiable? Where does

$$F'(x) = f(x)?$$

2. Repeat part (1) for the function

$$f(x) = \begin{cases} 1, & \text{if } x < 0, \\ 2, & \text{if } x \geq 0. \end{cases}$$

💡 Average Value Theorem

If g is continuous on $[a, b]$, show that there exists a point $c \in (a, b)$ such that

$$g(c) = \frac{1}{b-a} \int_a^b g(x) dx.$$

Theorem

Let

$$h(x) = \begin{cases} 1, & \text{if } x < 1 \text{ or } x > 1, \\ 0, & \text{if } x = 1, \end{cases}$$

and define

$$H(x) = \int_0^x h(t) dt.$$

Show that even though h is not continuous at $x = 1$, $H(x)$ is still differentiable at $x = 1$.

Theorem

The hypothesis in FTC-2 that

$$F'(x) = f(x)$$

for all $x \in [a, b]$ is slightly stronger than it needs to be. Carefully read the proof and state exactly what needs to be assumed with regard to the relationship between f and F for the proof to be valid.



💡 Updated Fundamental Theorem of Calculus Part 2

If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, and $F : [a, b] \rightarrow \mathbb{R}$ satisfies

$$F'(x) = f(x) \quad \text{for all } x \in (a, b),$$

then

$$\int_a^b f = F(b) - F(a).$$

Thank You!

We'd love your questions and feedback.

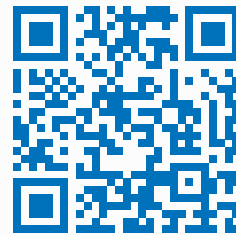
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(Lectures, walkthroughs, and course updates)



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References

- [1] Stephen Abbott, *Understanding Analysis*, 2nd Edition, Springer, 2015.
- [2] Terence Tao, *Analysis I*, 3rd Edition, Texts and Readings in Mathematics, Hindustan Book Agency, 2016.